Holding shorts to the time exit on the short-only book

Experiment report · L/S Gamma Desk, QUANTT

2026-10-16

Question: On the short-only book, does removing the flat-signal exit raise net P&L at 1x and 2x spreads, in both halves, without a worse daily tail?

Method:

- **Pre-registration:** the hypothesis card was committed before any run (commit dbf1d11).
- **Data:** the live paper rules (`configs/paper.toml`) are replayed on R2 replay snapshots v1 (config_hash b040760187f0) over the development window 2014-06-02 to 2024-12-31. The hold-out (2025-01-01 to 2026-07-10) was not used.
- **Book:** short-only (`[signal] long_band = -inf`): a short put near 30 DTE, 1 contract, hedged daily at 10:15 ET. A long signal never fires on this book, so there is no flip exit.
- **Variants:**
  - `short_only` (reference): the 3-day flat exit. Same config as `2026-10-16-short-only/short_only`.
  - `no_flat` (primary): `flat_days = 999`, so shorts close only at 7 DTE, on the stop (1x credit) or at the 50% profit-take.
  - `flat10` (sensitivity, reported only): `flat_days = 10`.
- **Runs:** every variant ran once at 1x and 2x spreads. Costs are $0.65 per contract and $0.005 per share, with a $1 minimum per order.

1 Result vs the pass bar

`no_flat` against `short_only` at the same spread multiple. P&L is net USD on daily equity for 1 contract. ES is the mean of the worst 5% of daily P&L.

Check	1x value	1x bar	2x value	2x bar	Result
Net P&L, full window	−676	> -774	−2,388	$> -2{,}351$	pass 1x, fail 2x
Net P&L, 2014-06 to 2019-12	−1,023	> -875	−1,540	$> -1{,}261$	fail
Net P&L, 2020-01 to 2024-12	+347	$> +101$	−849	$> -1{,}090$	pass
abs ES 5% $\leq 1.10 \times$ reference	91.2	≤ 94.1	99.8	≤ 101.7	pass
Event floor: time exits (1x)	76	≥ 20			met

Verdict: failed. Holding shorts to the time exit helps a little at 1x (+$98 over the full window) but loses at 2x (−$37). It is worse in 2014-2019 at both spreads. The book still loses money in every variant.

1.1 Full-window numbers (not part of the bar)

	short_only 1x	no_flat 1x	flat10 1x	short_only 2x	no_flat 2x	flat10 2x
Net P&L	−774	−676	−931	−2,351	−2,388	−2,514
Round-trip P&L before commissions	+318	+1,166	+766	−1,240	−536	−805
Sharpe on $250k	−0.15	−0.13	−0.18	−0.42	−0.45	−0.47
Max drawdown	2,220	1,816	2,009	3,222	3,355	3,534
ES 5% (daily)	−85.5	−91.2	−91.7	−92.4	−99.8	−99.3
Round trips	100	76	78	100	76	78
Average hold (days)	10.7	23.6	20.8			
Hedge orders (= hedge fees, USD)	770	1,188	1,097	770	1,188	1,097

2 Charts

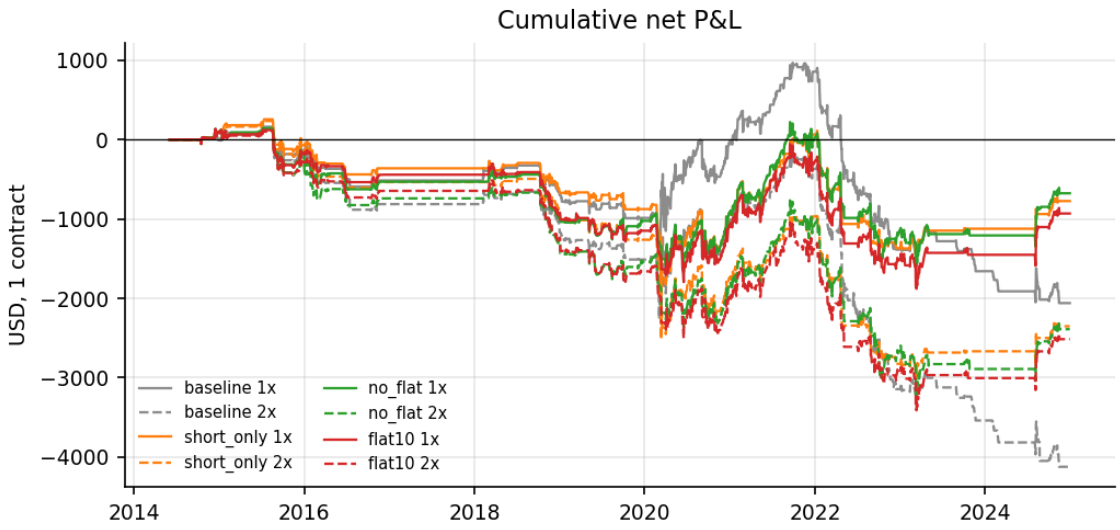


Figure 1: Cumulative net P&L, 1x solid and 2x dashed; baseline (live rules) grey

`no_flat` falls behind `short_only` in 2015-2016 and again in 2018, then catches up in 2019-2024. Most of the catch-up comes in 2022, where it loses $268 less at 1x. At 2x the lines end almost together.

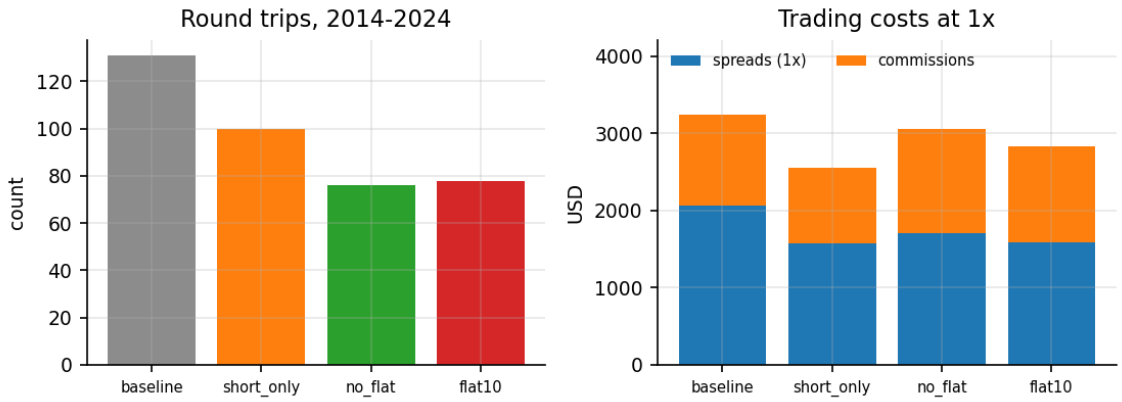


Figure 2: Round trips and trading costs at 1x

This chart shows the cost side. Round trips fall from 100 to 76, but total trading costs rise from about $2,550 to $3,050 at 1x: commissions are up $370 (hedges +$418, options −$48) and spreads are up $134.

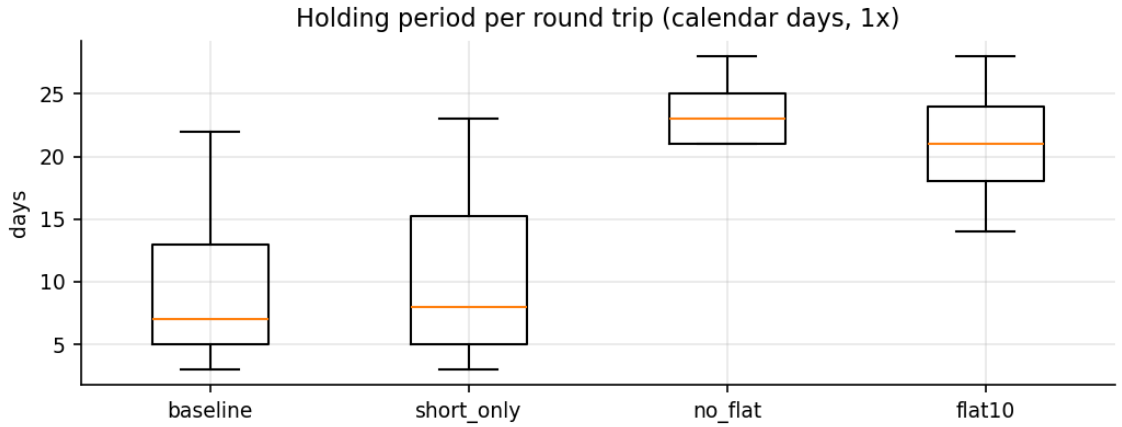


Figure 3: Holding period per round trip

Without the flat exit every short is held from entry near 30 DTE to the 7-DTE exit: median 23 days, 5th-95th percentile 21-28. `flat10` sits in between. The reference's median hold is 8 days.

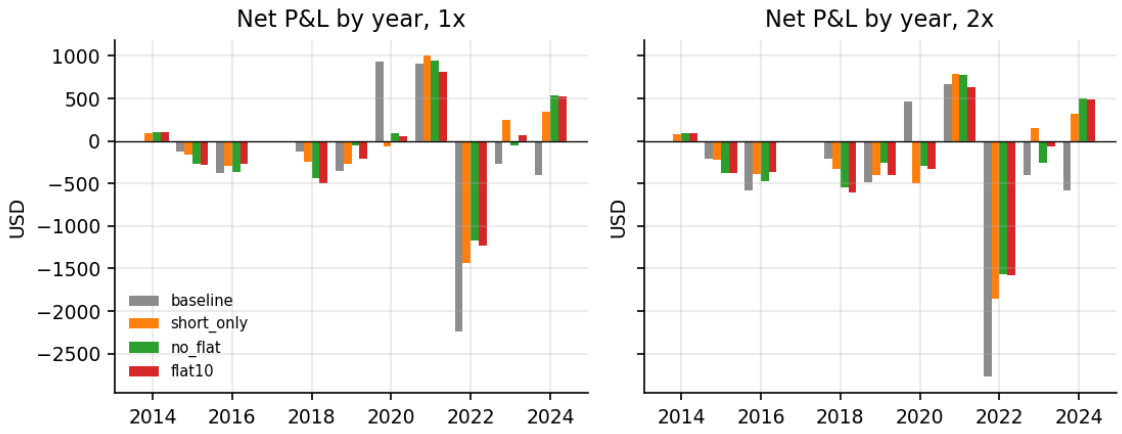


Figure 4: Net P&L by year

At 1x, holding longer helps in 2014, 2019 (−$56 vs −$270), 2020 (+$90 vs −$60), 2022 (−$1,165 vs −$1,433) and 2024 (+$531 vs +$349). It hurts in 2015-2016, 2018 (−$437 vs −$244), 2021 (+$948 vs +$1,000) and 2023 (−$57 vs +$244). The first-half shortfall comes from 2015-2016 and 2018 (−$381 together), partly offset by 2019 (+$214).

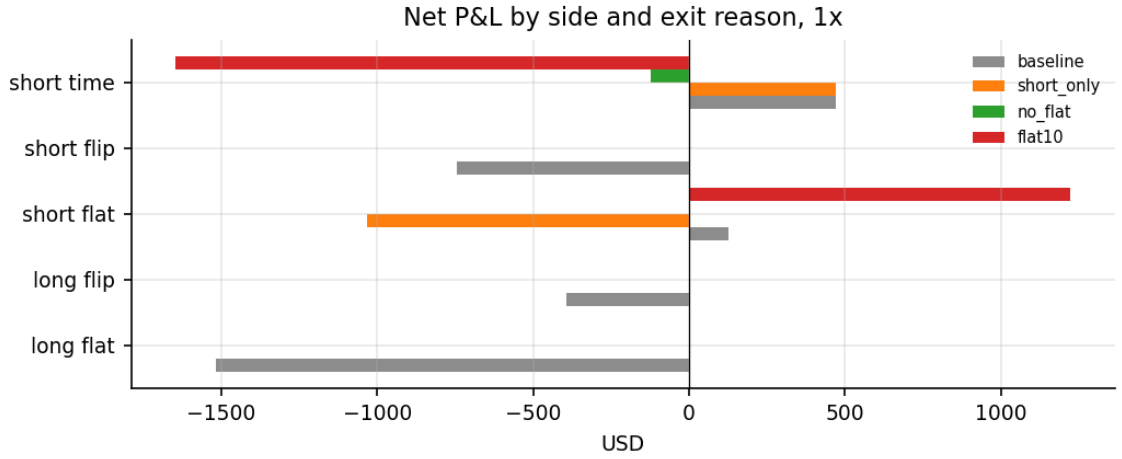


Figure 5: Net P&L by side and exit reason, 1x

Every `no_flat` trade closes at the time exit (no stop or profit-take fired): 76 trades, −$124 net in round trips. The reference's +$470 from 13 time exits did not scale. Holding every short to 7 DTE also holds the ones that are going wrong.

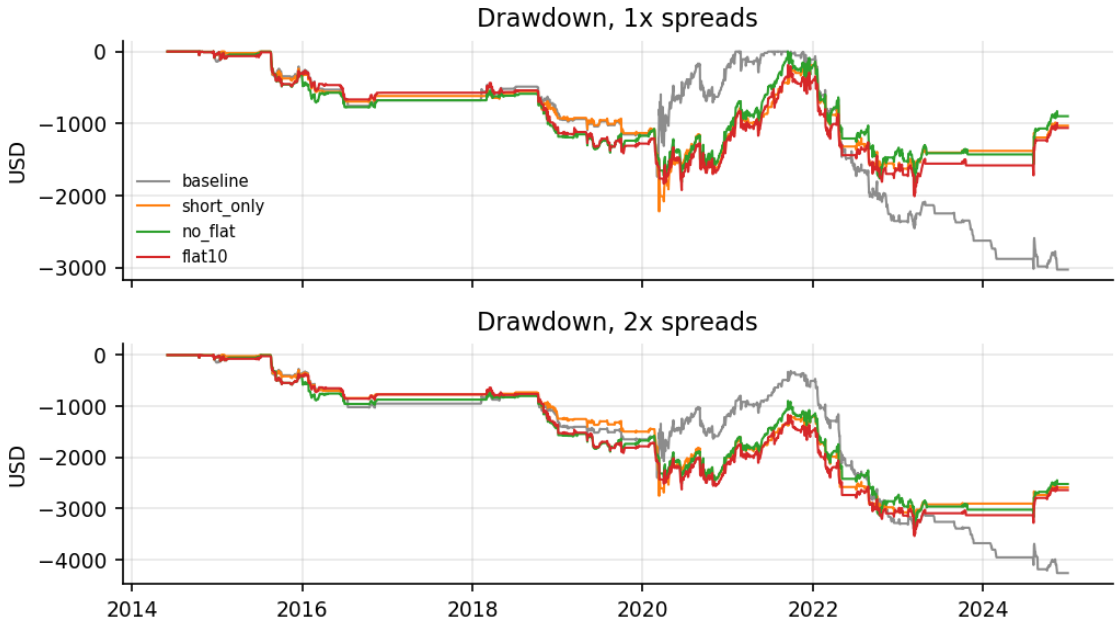


Figure 6: Drawdown at 1x and 2x spreads

The 1x maximum drawdown is smaller ($1,816 against $2,220). That is because the 2015-2020 decline was shallower, helped by 2019-2020. The reference's maximum drawdown runs from mid-2015 to the March 2020 trough. `no_flat`'s comes later (2021-09 to 2023-03), and over that stretch it is actually deeper than the reference's ($1,816 against $1,724). At 2x it is slightly larger overall.

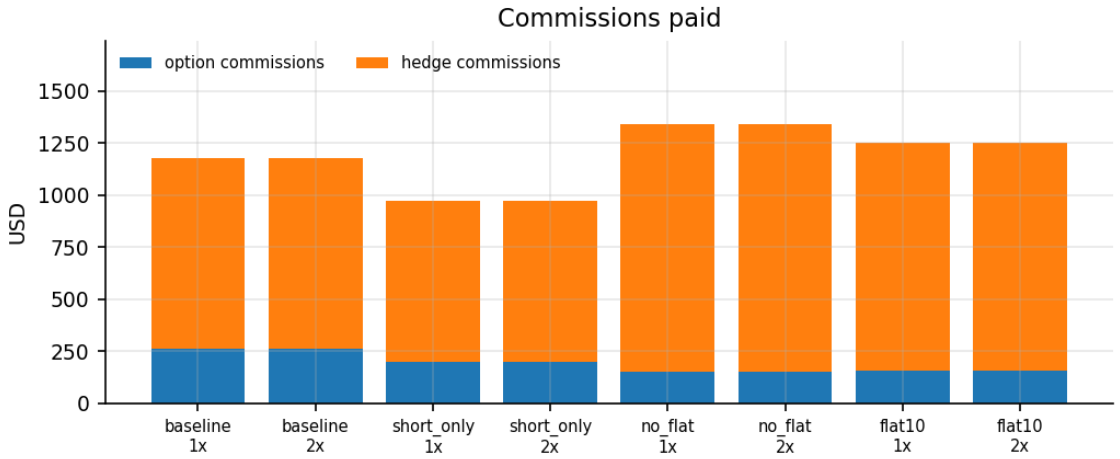


Figure 7: Commissions paid

Hedge commissions are most of the commissions in every short-only variant. Spreads (the turnover chart) are the larger cost overall.

3 Analysis

The hypothesis had two parts. Shorts would keep more of the late time decay, and they would pay fewer spreads. The first part held: round-trip P&L before commissions rose from +$318 to +$1,166 at 1x (3.7x). That figure is already net of the spreads paid inside each round trip. The second part did not hold. The 1x bridge from the reference to `no_flat` is: +$848 more round-trip P&L, −$410 more commissions attributed to round trips, −$340 more unattributed unwind cost, giving +$98. At 2x it is +$704 − $410 − $330 = −$37.

- **More hedge orders.** A short held for 24 days instead of 11 is hedged on more days. With 1 contract, each small daily hedge pays the $1 order minimum. Hedge orders rose from 770 to 1,188 (+$418), while shares traded fell about 6% (12,204 against 13,046 for the reference). So the extra cost is the fixed order minimum, not hedging volume. The 10-share hedge band from `2026-10-16-hedge-band` targets exactly this cost.
- **Costlier closes, already inside the round-trip P&L.** The option-leg spread cost rose from $1,512 to $1,651 (+$139) despite 24 fewer round trips. Holding to 7 DTE ends more shorts in the money (32% against 27%), and an in-the-money round trip costs far more to cross: about $56 against $38 for the reference's in-the-money round trips, and about $6-7 out of the money. The spread model fits a wider cost for deep-in-the-money options, and the 2x run doubles it. This cost is part of why the gross gain is $848 and not more.
- **Unwind cost, and most of it is noise.** About $552 at 1x is not attributed to any round trip, against $212 for the reference. It is the order that flattens the hedge after each close: a $1 commission plus the gap between the fill-window spot and the decision spot, times the shares. `no_flat` had fewer such orders, not bigger ones: 50 averaging 39 shares, against 90 averaging 44, because 25 of its closes re-entered at once and carried the shares over. The growth comes from the price gap: $502 against $122. A few days in 2020 and 2022 dominate (the five worst unwinds total more than the whole $502, and the median is about $0). The pass bar uses equity P&L, which includes it.

So the decay gain is real, and the flat exit was costing the book something. But on a 1-contract book with daily hedges, the extra $1 hedge orders take about half of the gain, and the unwind noise takes most of the rest. The 2x shortfall ($37) and the 1x gain ($98) are both smaller than the noise in a handful of unwind fills. The honest reading: on this cost structure, holding to the time exit is roughly neutral, not a clear improvement.

4 Verdict

Failed on the pre-registered bar: the 2x full window and the 2014-2019 half fail. Keep the live 3-day flat exit for now.

A follow-up worth pre-registering, which needs its own card and is in-sample: `no_flat` combined with the 10-share hedge band, which cuts the cost that grew here. Its reference should be short-only plus the 10-share hedge band (queued as "short-only + hedge band").

5 Caveats

- **In-sample:** the idea came from the baseline's exit-reason breakdown on this same window.
- **One contract:** the $1 order minimum turns longer holds into many small hedge orders. With more contracts per position the hedge orders would be larger and that cost would matter less.
- **Spread model:** fitted on about two weeks of calm 2026 live quotes. The in-the-money spread tier drives part of this result, and the 2x run is the stress test.
- **Data gaps:** R2 has no month-end weekdays, so 89 scheduled times (3.3%) are skipped in every variant.

- **Few events per year:** 31 trades in 2014-2019 and 45 in 2020-2024 for `no_flat`. The yearly differences rest on a handful of trades each.
- **Unwind noise:** the hedge-flattening orders fill on a spot price 15-30 minutes after the decision spot, and a few such fills are larger than the margins in this test.

6 Trials

6 trials for this experiment (3 variants at 1x and 2x, one run of `experiment.py`), plus the baseline's 2. `short_only` repeats the config of `2026-10-16-short-only/short_only` (2 earlier trials). Its numbers here are identical: −$773.8 at 1x, −$2,351.3 at 2x. So across the desk the same short-only test is registered 4 times. The pass bar was not changed.